

## OFFICE NETWORK BUILD

### Trunking and Inter-VLAN Routing

In the previous section, VLANs were implemented to logically separate the office departments into distinct Layer 2 broadcast domains. This allowed Tech, Business Development, Finance, and Admin to operate within their own VLANs, preventing broadcast traffic from unnecessarily spreading across other departments.

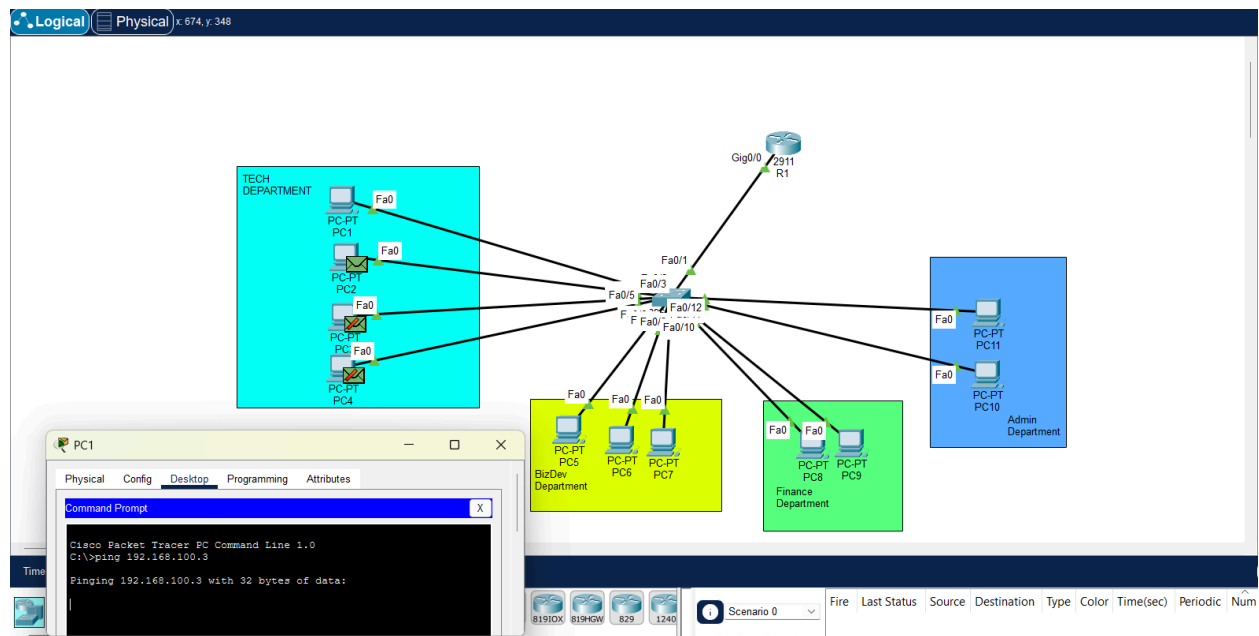
However, this segmentation also creates a new requirement: devices in different VLANs cannot communicate directly at Layer 2. In a real office environment, there may be legitimate situations where one department needs to access resources or services located in another VLAN. To address this, this section introduces trunking and inter-VLAN routing.

Trunking allows traffic from multiple VLANs to travel across a single physical link between the switch and router, while inter-VLAN routing allows the router to route traffic between the separate VLANs when communication is necessary. Together, these technologies maintain the logical separation established in the previous VLAN section while still allowing controlled communication between departments.

Shown below is the current IP Address table with their Default Gateways and VLANs included:

| Office Network Build |                |                       |                 |         |           |
|----------------------|----------------|-----------------------|-----------------|---------|-----------|
| Device               | IP             | Subnet                | Default Gateway | VLAN    | Interface |
| R1                   | N/A            | N/A                   | N/A             | N/A     | G0/0      |
| PC1                  | 192.168.100.2  | 255.255.255.248 - /29 | 192.168.100.1   | VLAN 10 | Fa0       |
| PC2                  | 192.168.100.3  | 255.255.255.248 - /29 | 192.168.100.1   | VLAN 10 | Fa0       |
| PC3                  | 192.168.100.4  | 255.255.255.248 - /29 | 192.168.100.1   | VLAN 10 | Fa0       |
| PC4                  | 192.168.100.5  | 255.255.255.248 - /29 | 192.168.100.1   | VLAN 10 | Fa0       |
| PC5                  | 192.168.100.10 | 255.255.255.248 - /29 | 192.168.100.9   | VLAN 20 | Fa0       |
| PC6                  | 192.168.100.11 | 255.255.255.248 - /29 | 192.168.100.9   | VLAN 20 | Fa0       |
| PC7                  | 192.168.100.12 | 255.255.255.248 - /29 | 192.168.100.9   | VLAN 20 | Fa0       |
| PC8                  | 192.168.100.18 | 255.255.255.248 - /29 | 192.168.100.17  | VLAN 30 | Fa0       |
| PC9                  | 192.168.100.19 | 255.255.255.248 - /29 | 192.168.100.17  | VLAN 30 | Fa0       |
| PC10                 | 192.168.100.26 | 255.255.255.248 - /29 | 192.168.100.25  | VLAN 40 | Fa0       |
| PC11                 | 192.168.100.27 | 255.255.255.248 - /29 | 192.168.100.25  | VLAN 40 | Fa0       |

## Current Situation:



The image above shows the current state of the network after implementing VLANs. When a PC communicates with another device within the same VLAN for the first time, it uses ARP to discover the destination MAC address. Because the network has been segmented into VLANs, the ARP request is now contained within its assigned VLAN rather than being broadcast across the entire Layer 2 network. This provides practical evidence that our VLAN implementation is successfully separating the departments into distinct broadcast domains.

However, this segmentation introduces another requirement: how can a device in one VLAN communicate with a device in another VLAN when communication is necessary? To address this, the next stage of the project introduces trunking and inter-VLAN routing. Trunking will allow multiple VLANs to traverse the single link between the switch and router, while inter-VLAN routing will enable communication between the otherwise separate VLANs.

## Trunking

Since SW1 is connected to the router via interface fa0/1, this interface was configured as a Trunk with 802.1Q VLAN tagging:

```
SW1
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport mode trunk
!
interface FastEthernet0/2
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/3
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/4
```

## Inter-VLAN Routing/Router-on-a-Stick

After configuring the trunk, the sub-interfaces in the router were configured to respond to the VLANs via the Trunk line.

```
R1
IOS Command Line Interface
!
!
!
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.100.1 255.255.255.248
!
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.100.9 255.255.255.248
!
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 192.168.100.17 255.255.255.248
!
interface GigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 192.168.100.25 255.255.255.248
!
interface GigabitEthernet0/1
 no ip address
!

R1# show ip interface brief

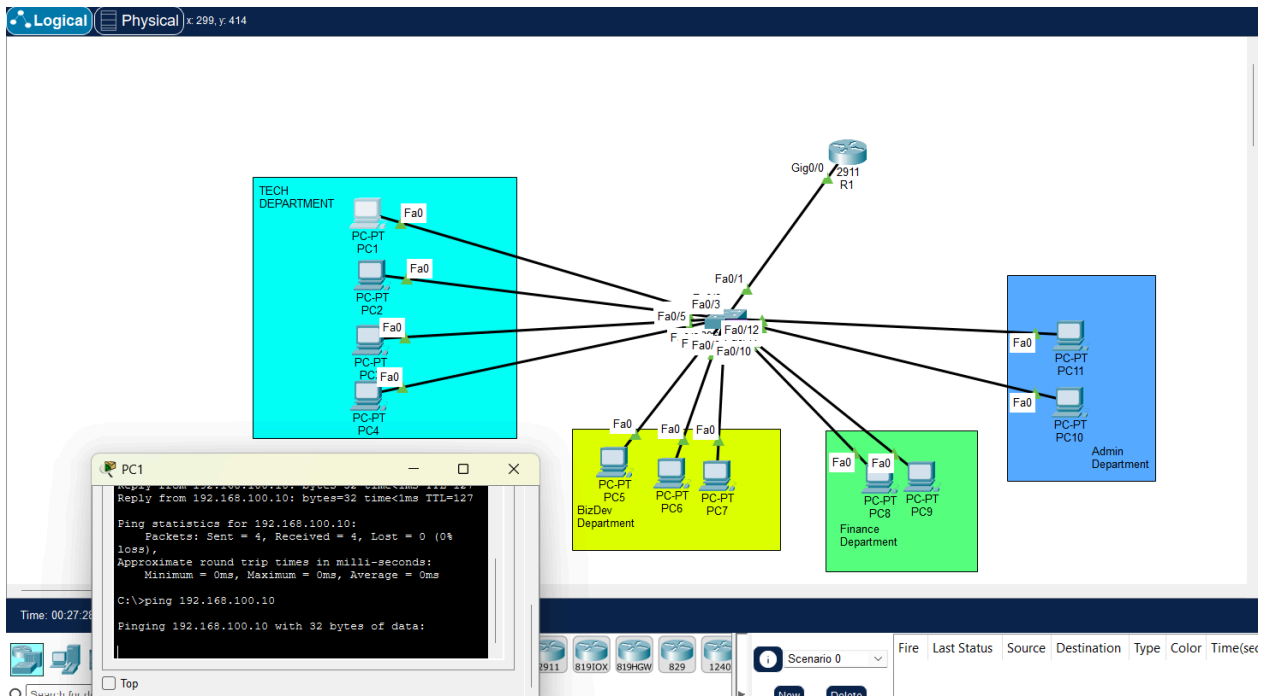
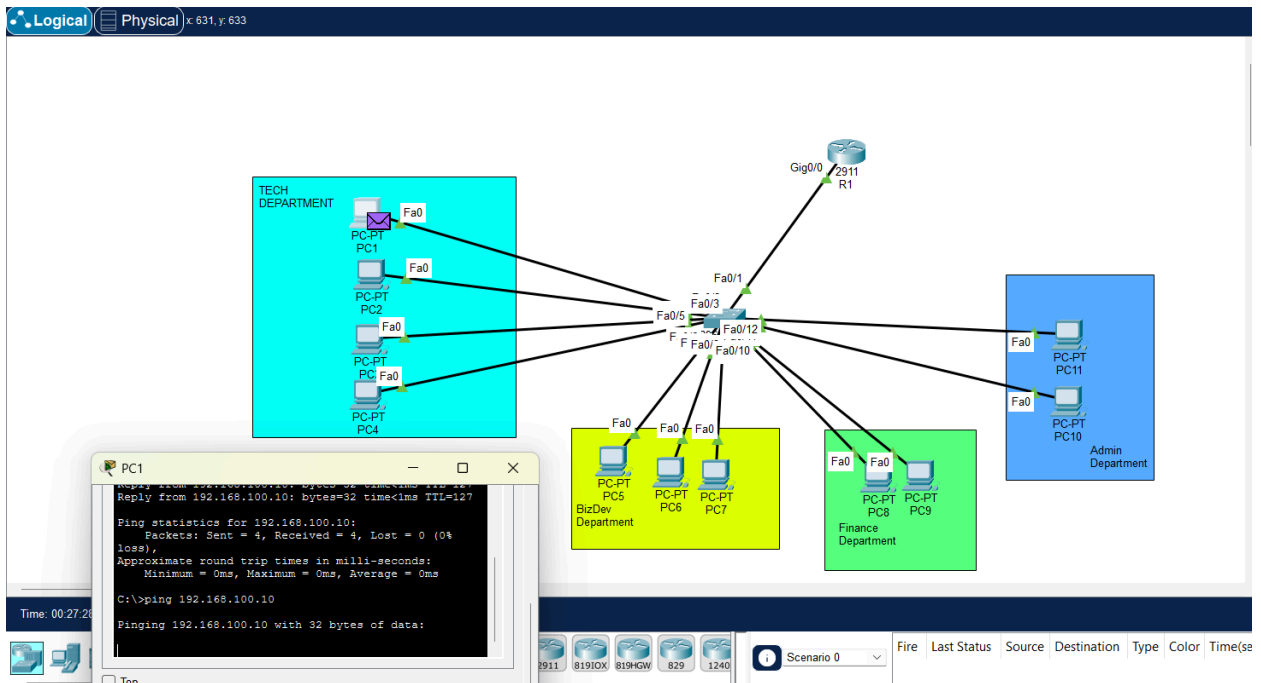
```

| Interface             | IP-Address     | OK? | Method | Status                | Protocol |
|-----------------------|----------------|-----|--------|-----------------------|----------|
| GigabitEthernet0/0    | unassigned     | YES | unset  | up                    | up       |
| GigabitEthernet0/0.10 | 192.168.100.1  | YES | manual | up                    | up       |
| GigabitEthernet0/0.20 | 192.168.100.9  | YES | manual | up                    | up       |
| GigabitEthernet0/0.30 | 192.168.100.17 | YES | manual | up                    | up       |
| GigabitEthernet0/0.40 | 192.168.100.25 | YES | manual | up                    | up       |
| GigabitEthernet0/1    | unassigned     | YES | unset  | administratively down | down     |
| GigabitEthernet0/2    | unassigned     | YES | unset  | administratively down | down     |
| Vlan1                 | unassigned     | YES | unset  | administratively down | down     |

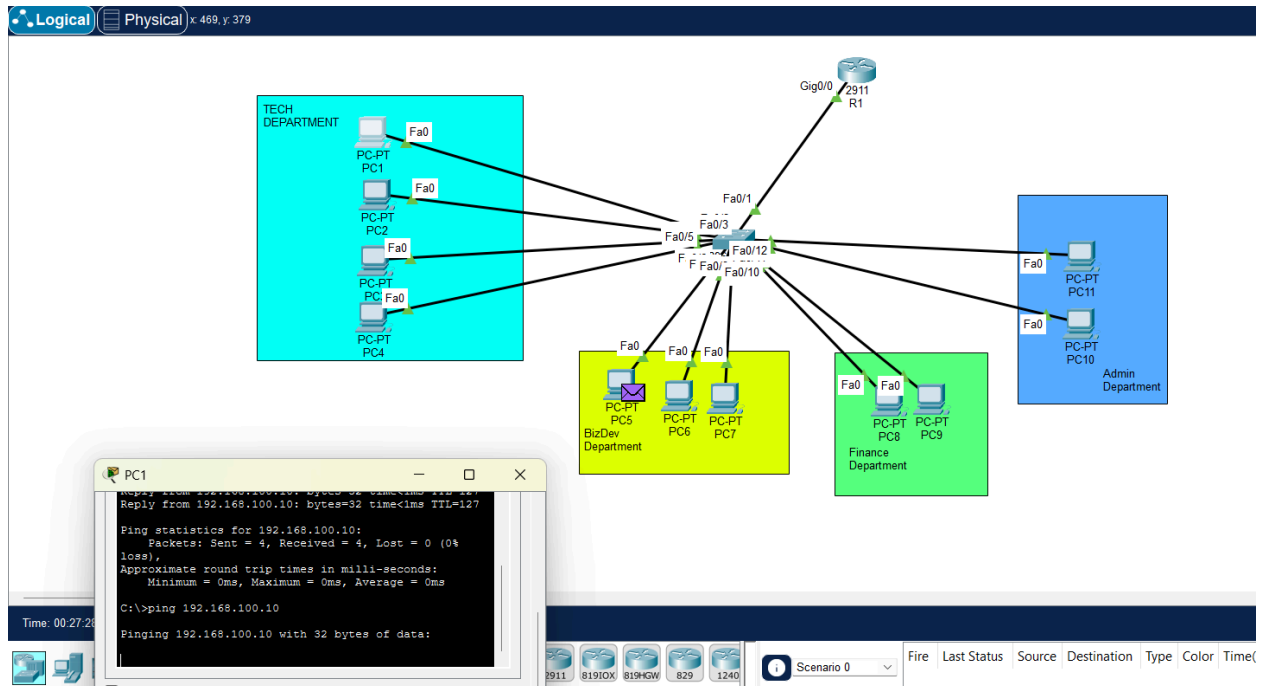
```
R1#
```

## Ping Tests

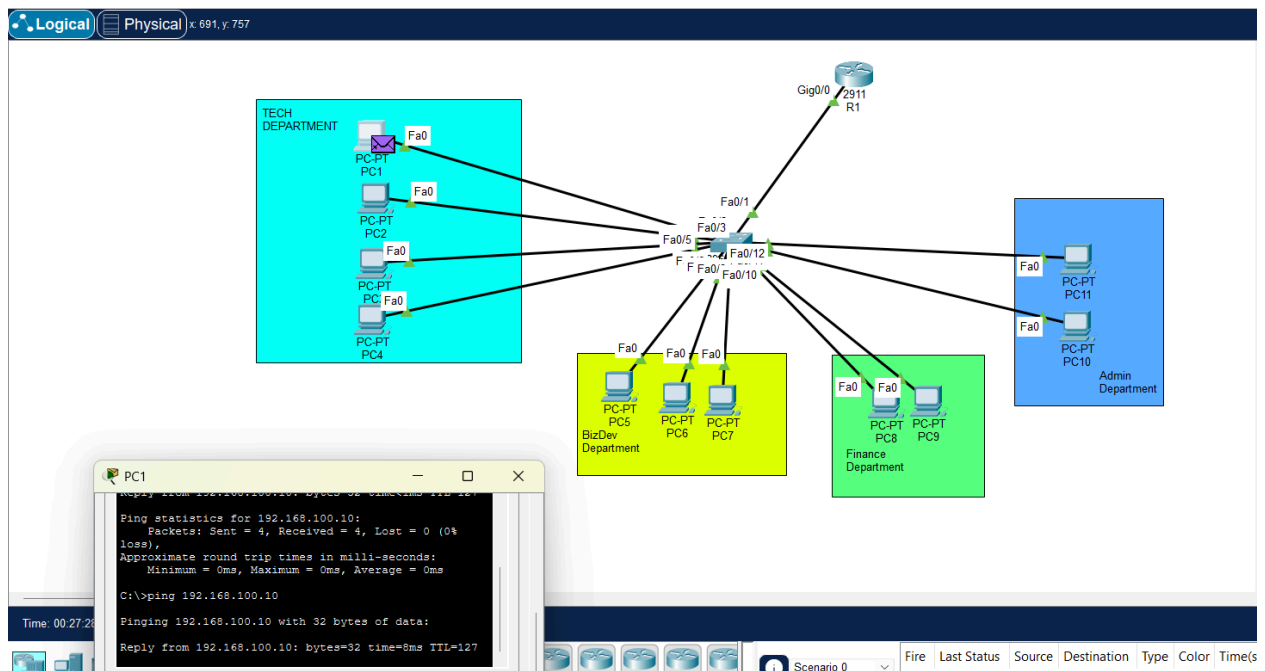
### PC1 (Tech Department - VLAN 10) to PC5 (BizDev Dept - VLAN 20)







## Return:



**Note:** Only 1 test was documented in this file, but overall connectivity was tested. Feel free to download the .pkt file to test!

**Conclusion:**

The implementation of trunking and inter-VLAN routing successfully enabled communication between the previously isolated departmental VLANs while preserving their Layer 2 segmentation. The switch-to-router link was configured as an 802.1Q trunk to carry traffic from multiple VLANs over a single physical connection, while Router-on-a-Stick provided a dedicated gateway for each VLAN and routed traffic between their respective subnets.

Testing confirmed that devices within the same VLAN could communicate normally, while devices in different VLANs could now communicate through the router when necessary. This configuration maintained the benefits of VLAN-based segmentation while providing the connectivity required for normal business operations.